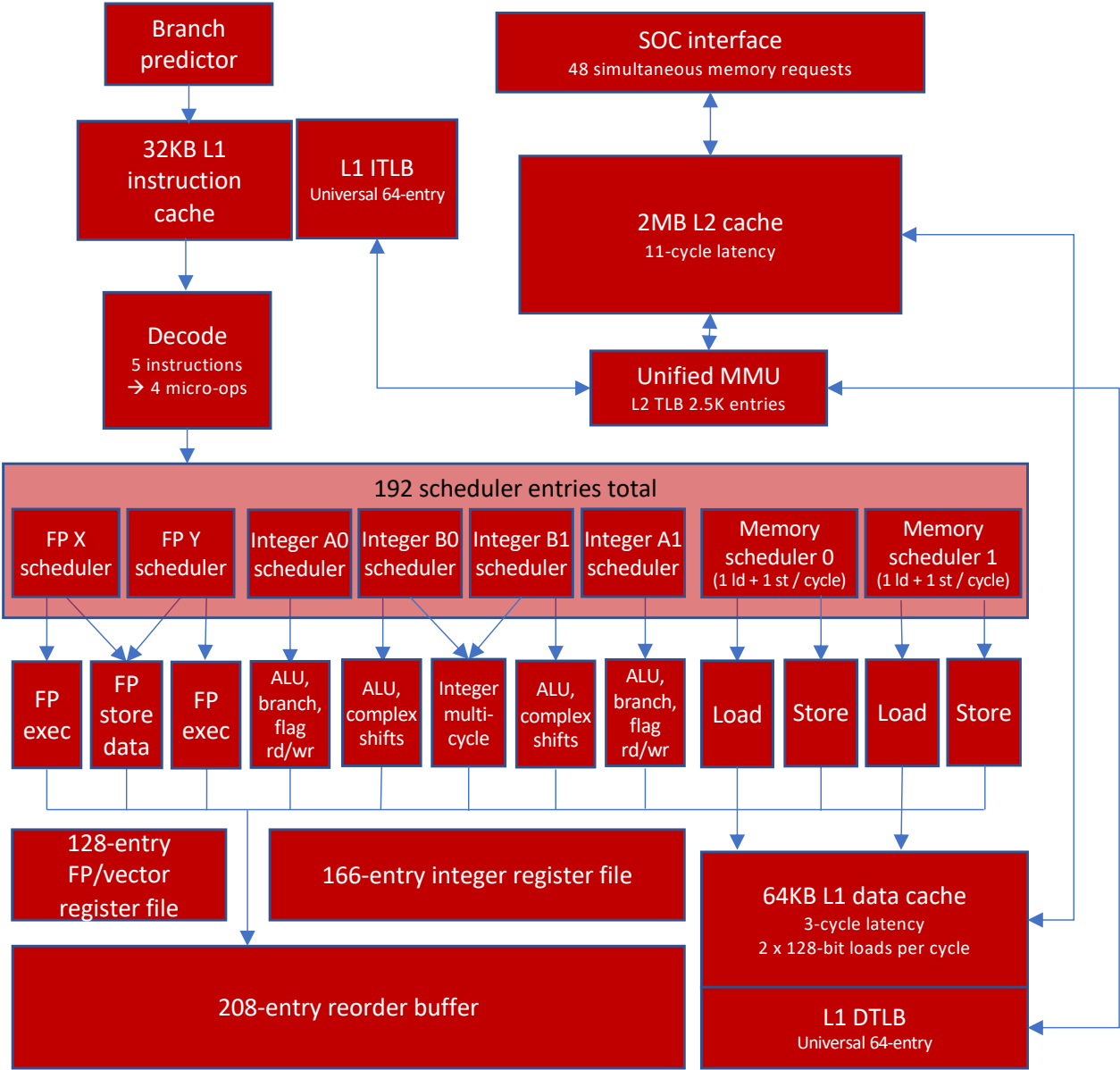

Branch predictor
SOC interface
48 simultaneous memory requests
32KB L1 instruction cache
L1 ITLB
Universal 64-entry
2MB L2 cache
11-cycle latency
Decode
5 instructions
→ 4 micro-ops
Unified MMU
L2 TLB 2.5K entries
192 scheduler entries total
FP X scheduler
FP Y scheduler
Integer A0 scheduler
Integer B0 scheduler
Integer B1 scheduler
Integer A1 scheduler
Memory scheduler 0
(1 ld + 1 st / cycle)
Memory scheduler 1
(1 ld + 1 st / cycle)
FP exec
FP store data
FP exec
ALU, branch, flag rd/wr
ALU, complex shifts
Integer multi-cycle
ALU, complex shifts
ALU, branch, flag rd/wr
Load
Store
Load
Store
128-entry FP/vector register file
166-entry integer register file
64KB L1 data cache
3-cycle latency
2 x 128-bit loads per cycle
208-entry reorder buffer
L1 DTLB
Universal 64-entry